MUSCULOSKELETAL DIGITAL TWIN · INDEPENDENT STUDY

Technical Review of LB-AO

Critique of current work and a proposed improvement plan

Scope: LB-AO v0.5 (hip ontology v0.2.0, LB-QPS v0.3.0, gluteus maximus dataset v0.2.0)

Prepared for Corbin Wright and collaborators · October 7, 2026

Executive summary

The project has a strong core idea: keep anatomical fact, biomechanical convention, measured value, model parameter and model assumption as separate, traceable things. The gluteus maximus reference dataset shows this working well. However, the software does not yet enforce the rules the design documents describe, the anatomy and numbers layers disagree on several IDs and class names, and nothing simulates movement yet.

18 / 18 existing tests pass	**8 / 8** deliberately invalid records accepted by the validator	**0** lines of dynamics or simulation code

Top findings

- **Cross-file IDs don't match.** Gluteus maximus records point to a joint coordinate that doesn't exist, and a parameter name differs between the ontology and the schema.
- **Class names don't line up**, so the hip joint, labrum and cartilage receive no numeric properties.
- **The validator accepts almost anything**: wrong units, negative mass, unknown sources, and any value marked “assumed” with no justification.
- **The anatomy layer doesn't yet meet the project's own evidence standard** (citations are per entity, not per claim).
- **The clinical purpose of the project has no implementation yet**: no dynamics, no condition model, no personalization.

RECOMMENDATION

Fix data integrity first (about a week of focused work), then build a thin end-to-end planar squat simulation that the ontology feeds. Let the simulation's needs drive further ontology detail, rather than expanding anatomy ahead of use.

SECTION 1

Scope and method

All 18 uploaded files were reviewed. The code was rebuilt into the folder layout the tests expect and run directly; findings marked *verified* were confirmed by executing probe scripts against the code and data, not just by reading them.

Group	Files	Role
Design documents	Framework PDF; LB-AO v0.1 Ontology PDF; Source & Evidence Standard PDF; QPS Rationale PDF	Concept, entity model, evidence policy, quantitative rationale
Data	`hip.yaml`, `quantitative_schema.yaml`, `gluteus_maximus.yaml`	Anatomy ontology, property schema, reference muscle dataset
Code	`src/ontology.py`, `src/quantitative.py`, `__init__.py`	Loading, querying and validation
Tests and demos	2 test files (18 tests); 3 demo scripts; README; requirements	Behavior checks and walkthroughs
Not reviewed	2 `.pyc` files	Compiled bytecode; should not be shared

Where the project stands

Conceptual framework	Framework PDF — entities, relationships, twin loop	**IN PLACE**
Anatomy ontology	hip.yaml — 171 entities, 22 muscles/side, 3 hip DOFs	**IN PLACE**
Quantitative property schema	LB-QPS + gluteus maximus reference dataset	**IN PLACE**
Segment dynamics / simulation	Equations of motion, integrator, squat task	**NOT STARTED**
Condition → parameter change	Conditions as evidence-backed parameter changes	**NOT STARTED**
Measurement & personalization	Clinic measures → parameter estimation	**NOT STARTED**

Layers ordered from foundation (top) to the clinical use case (bottom).

The first three layers are in place. The framework document's first concrete goal, a sagittal squat with ankle, knee, hip and trunk segments, depends on the fourth layer, which has not been started. The project's stated purpose, tuning parameters to represent clinical conditions, depends on the fifth.

SECTION 2

What is working well

- **Fact vs. assumption is a first-class idea.** “Anatomical fact ≠ biomechanical convention ≠ measured quantity ≠ estimated parameter ≠ model assumption” is the right foundation for a clinical tool. Most student models never make this distinction.
- **Property roles.** Separating *parameters* (optimal fiber length), *functions of state* (moment arm r(q)) and *state outputs* (muscle force F(t)) prevents a common modeling error.
- **Honest gaps.** Tendon slack length, contraction velocity and activation constants are left unresolved instead of filled with convenient numbers.
- **Derivations are preserved.** Maximum isometric force keeps its PCSA input and the 61 N/cm² specific-tension assumption. The arithmetic checks out (30.37 cm² × 61 N/cm² = 1852.6 N).
- **Symmetry treated as a reversible assumption** rather than an anatomical claim, with a stated override rule for future personalization.
- **Uncertainty survives unit conversion**, and every conversion is recorded.
- **A clear simulation abstraction.** The `squat_v0_1` model block records which coordinates are retained and omitted and why.

SECTION 3

Defects (verified)

Each item below was reproduced by running code. Severity reflects how much the issue would compromise results or block growth as more muscles and joints are added.

#	Issue	Severity	Evidence and impact
D1	**Dangling joint-coordinate ID**	High	Moment-arm and musculotendon-length records reference `LB:DOF:HIP_FLEXION_R`; the ontology defines `LB:DOF:HIP_FLEX_EXT_R`. No check runs across files, so this class of error will multiply with each new muscle.
D2	**Parameter name mismatch**	Medium	Every muscle in `hip.yaml` lists `pennation_angle`; the schema defines `pennation_angle_at_optimal`.
D3	**Class names don't match the schema**	High	Schema targets `SynovialJoint`, `Labrum`, `Cartilage`; the ontology uses `SynovialBallAndSocketJoint`, `Fibrocartilage`, `ArticularCartilage`. Eleven ontology classes receive no quantitative properties, including the hip joint itself.
D4	**Validator accepts invalid records**	High	All 8 deliberately bad records passed with zero errors (see Appendix A). Most serious: `datum_kind: assumed` bypasses the evidence requirement entirely, and a status typo (`Populated`) skips validation altogether.
D5	**Schema contradicts itself and the data**	Medium	`musculotendon_length` has role `function_of_state` but form `scalar`. `muscle_force` is `scalar` in the schema but `time_series` in the data. Records are never checked against their property definitions.

#	Issue	Severity	Evidence and impact
D6	**Shared defaults misattribute applicability**	Medium	The `common` block stamps Ward 2009's cadaver cohort (n=18, elderly) onto every record, including the Arnold 2010 model-derived force and the unresolved model-specific values.
D7	**Source conflict not recorded**	Medium	Ward reports PCSA 33.4 cm²; Arnold's model uses 30.37 cm². That is 2037 N vs. 1853 N of maximum force, yet `conflicts_with` is empty, contrary to the schema's own rule.
D8	**Laterality bug in bilateral records**	Low	The moment-arm record targets left and right muscles but its coordinate context points only to the right hip.

SECTION 4

Gaps against the project's own standards

These are places where the implementation does not yet do what the design documents promise.

Stated principle	Current state	Suggested remedy
Evidence supports specific claims, not whole entities (Evidence Standard §8)	Ontology attaches StatPearls to whole entities; only the numbers layer is claim-specific.	Move ontology evidence to per-relationship claims, e.g. a source per `INNERVATED_BY` edge.
Laterality is explicit (ontology principle)	Nerves have no side; left and right gluteus maximus share one inferior gluteal nerve entity.	Split nerves by side (or model nerve roots as midline and branches as sided).
Deferred cross-module attachments are documented (ontology principle)	Nine muscles per side lack an origin or insertion (psoas, hamstrings, rectus femoris, sartorius, gracilis, piriformis, obturator internus). Nothing records why.	Add a `deferred_attachments` list naming the target module (lumbar spine, knee, sacrum).
Controlled relationship vocabulary	8 defined predicates are never used (incl. `IS_A`, `COVERED_BY`). PDF predicates `SURROUNDS` and `REINFORCED_BY` are missing.	Use `IS_A` to build a class hierarchy (also fixes D3). Reconcile vocabulary with the PDF.
Consistent bidirectional structure	73 `PART_OF` links have no matching `HAS_PART`.	Store one direction and derive the inverse in code.
Backward compatibility and versioning (QPS Rationale §11)	README says v0.5; hip.yaml 0.2.0; schema 0.3.0; PDFs v0.1. No changelog.	Single project version, `CHANGELOG.md`, and git tags per release.

SECTION 5

Anatomical accuracy and completeness

- **Adductor magnus** has dual innervation: obturator nerve (adductor part) and tibial division of the sciatic nerve (hamstring part). Only the obturator is listed. This matters for any nerve-injury condition and argues for modeling the muscle as two compartments.
- **Gluteus maximus origins** also include the coccyx, sacrotuberous ligament and thoracolumbar fascia.

- **Gluteus medius and minimus** use the generic gluteal surface, although the anterior, posterior and inferior gluteal lines are already defined as landmarks and would give their true origin boundaries.
- **Biarticular muscles** (rectus femoris, hamstrings, gracilis, sartorius) list knee actions but cross only the hip. The knee will be essential for the squat.
- **Items in the v0.1 PDF but not in the YAML:** sacroiliac, sacrotuberous and sacrospinous ligaments; lesser sciatic notch; `LB:FRAME:FEMUR`.
- **Prefix inconsistency:** the iliotibial tract uses the `TEN` prefix but class `FascialThickening`.
- **To confirm against Ward 2009, Table 3:** whether the 15.69 cm fiber length is already normalized to optimal sarcomere length. The “optimal fiber length” label depends on it.

SECTION 6

The main strategic risk

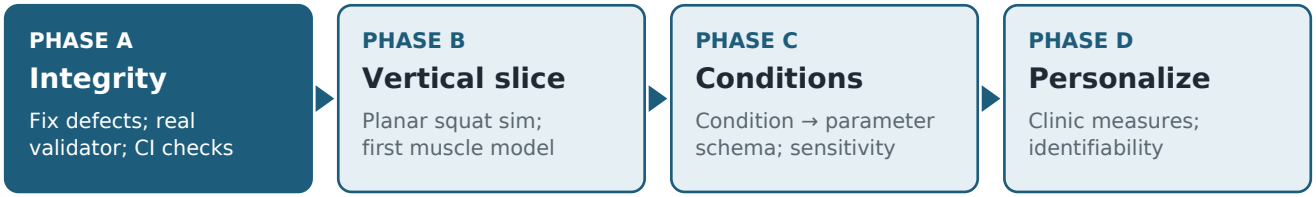
RISK

Roughly 2,300 lines of hip anatomy have been written, with no dynamics. Expanding the ontology ahead of a working simulation risks building detail that the model never uses, and leaves the clinical purpose untested.

The framework document's first target needs, at minimum, mass, length, center-of-mass and inertia for foot, shank, thigh and head-arms-trunk segments, plus an equation of motion $M(q)q'' + C(q,q') + G(q) = \tau$. None of that exists yet. A thin working simulation would show which ontology fields are actually needed, and is where a dynamics and control-theory background contributes most.

SECTION 7

Improvement plan



Phase A: Data integrity

Goal: make the existing rules enforceable before adding more data.

- Fix D1–D8 and the standards gaps in Section 4.
- Replace the hand-written validator with a declarative schema (Pydantic or LinkML). This closes most loopholes by construction: enums for status and datum kind, required non-null fields, typed values.
- Check units with **pint**: each datum's unit must convert to its property's canonical unit.
- Add a cross-file referential check: every target, frame, coordinate and source ID must resolve.

- Require a written justification for any `assumed` value.
- Add one failing-case test per loophole in Appendix A; run the full suite in CI on every change.
- Merge the three copies of the defaults-merging logic into one function.

Done when: every row of Appendix A is rejected with a clear message, and all cross-file IDs resolve.

Phase B: Vertical slice

Goal: a planar squat that runs end to end on ontology data.

- Add segment-inertial data (de Leva 1996 is the usual source) using the existing schema.
- Build a 3-DOF planar model (ankle, knee, hip) driven by joint torques; integrate and visualize.
- Add the knee and shank to the ontology only as far as the model needs.
- Add gluteus maximus as the first Hill-type muscle; resolve its model-specific parameters by choosing a specific formulation (e.g. Thelen 2003 or De Groote 2016) and citing it.
- Generate the model from the ontology, so a change to a datum changes the simulation.

Done when: changing a gluteus maximus parameter in YAML visibly changes the simulated squat.

Phase C: Conditions

Goal: make the clinical use case concrete.

- Define a `Condition` → `ParameterModification` schema with the same evidence rigor as other data (e.g. a condition scales Fmax by a cited factor, with applicability).
- Start with two or three well-studied cases (e.g. hip abductor weakness, reduced hip ROM).
- Use the stored uncertainties for Monte Carlo sensitivity analysis. Note: pennation SD (26°) exceeds its mean (22°), so it needs a truncated or log-normal distribution, not a normal one.

Done when: a condition can be toggled and its effect on squat outcomes reported with uncertainty bands.

Phase D: Personalization

Goal: connect the model to measurements a clinic can actually take.

- Map clinical measures (manual muscle testing grades, goniometric ROM, dynamometry, video kinematics) to model parameters and outputs.
- Study **parameter identifiability**: which parameters can be estimated from those measures, and which cannot. This is a natural controls-theory contribution and a strong independent-study topic.
- Implement the override precedence the README describes: subject-specific > side-specific > shared baseline.

Done when: a set of clinic measurements produces a personalized parameter set with stated confidence.

SECTION 8

Ideas worth developing further

Idea	Why it's valuable
Link to established ontologies (FMA, Uberon)	Adding cross-reference IDs makes the work interoperable and lets reviewers check anatomy against a standard.
OpenSim export	Generating an `.osim` model lets results be compared against a widely validated tool.
Muscle compartments	Gluteus maximus, gluteus medius and adductor magnus all have region-dependent actions; the data already hints at this (`rotation_depends_on_fiber_region`).
Teaching mode for PT students	Toggle a condition, watch the squat change, see the traced evidence. This turns the provenance work into a visible learning feature.
Provenance viewer	Answer “what number did the simulation use, and why?” from the UI. The QPS Rationale already frames this question well.

Minor cleanup

- Remove the dead transformation-history branch in `validate_datum`.
- Strengthen `test_muscle_inventory` (currently only checks ≥ 20 muscles); assert the exact list.
- Add a `.gitignore` for `__pycache__` and `.pyc` files.
- Put the project under git with tagged releases, matching the versioning principle.

SECTION A

Validator probe results

Each row starts from the valid gluteus maximus mass record and introduces one fault. “Accepted” means the validator returned no errors for the faulty record.

Fault introduced	Expected	Actual result
Unit changed to `lb` (value in kg)	Reject	**Accepted** (no errors)
Unit set to null	Reject	**Accepted** (no errors)
Value set to −5 kg	Reject	**Accepted** (no errors)
Status `Populated` (capital P), no value, no source	Reject	**Accepted** (no errors)
`datum_kind: assumed`, value 9999, no source or rationale	Reject	**Accepted** (no errors)
Applicability level `individual-ish`	Reject	**Accepted** (no errors)
Source ID that does not exist	Reject	**Accepted** (no errors)
Target muscle that does not exist	Reject	**Accepted** (no errors)

Other verified checks

Check	Result
Existing test suite	18 / 18 pass
Quantitative IDs that resolve in the ontology	2 coordinate references fail (D1)
Ontology classes with no quantitative properties	11 of 19
Defined predicates never used	8 of 25
`PART_OF` links without inverse `HAS_PART`	73
Fmax arithmetic (30.37 cm² × 61 N/cm²)	1852.57 N — matches stored 1852.6 N

References

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- Arnold EM, Ward SR, Lieber RL, Delp SL. A model of the lower limb for analysis of human movement. *Ann Biomed Eng.* 2010;38(2):269–279.
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